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SAS project: CYBERSECURITY

Final Report

Date: 28-04-2025

```
*****
=====Author:Rashpinder Kaur Chahal=====
=====SAS PROJECT=====
=====Date:29 April 2025=====
The business objective is to enhance the organization's cybersecurity defense strategy
by understanding patterns in network attacks and identifying key characteristics of
successful intrusions vs. blocked ones. This will help in allocating security resources
more effectively and improving automated threat detection systems.....
*****;
```

```

PROC IMPORT DATAFILE="D:\sas\Sas project\cybersecurity_final.csv"
    OUT=SAS.cyber
    DBMS=CSV
    REPLACE;
    GETNAMES=YES;
    DATAROW=2;
RUN;

```

***Display records;**

```

proc print data=SAS.cyber (obs=15);run;

```

*

Columns Description:

timestamp Date and time of the event
source_ip IP address initiating the connection/attack
destination_ip Target IP address
protocol Network protocol used (HTTP, HTTPS, ICMP, etc.)
attack_type Type of cyberattack (e.g., Ransomware, SQL Injection)
severity Severity level (Low, Medium, High)
status Whether the attack was Blocked or Successful
packet_size Size of the packets involved
duration Duration of the event in seconds
num_packets Number of packets transferred
attack_score Calculated risk or threat score (possibly from a model or rule engine)
;

Obs	timestamp	source_ip	destination_ip	protocol	attack_type	severity	status	packet_size	duration	num_packets	attack_score
1	01JAN24:00:00:00	129.153.198.180	166.25.139.113	HTTP	Ransomware	Low	Blocked	952	118.78	440	2432.72094
2	01JAN24:01:00:00	183.214.39.114	38.64.254.108	HTTPS	SQL Injection	High	Blocked	503	89.67	40	1105.1151629
3	01JAN24:02:00:00	113.163.209.250	48.74.5.250	ICMP	Phishing	Low	Blocked	489	260.1	363	3312.0586475
4	01JAN24:03:00:00	133.144.103.224	23.112.190.220	HTTPS	DDoS	High	Successful	1394	573.24751623	280	6050.0634842
5	01JAN24:04:00:00	130.8.186.60	210.174.162.53	HTTPS	Malware	High	Successful	1316	257.34	226	3147.9268327
6	01JAN24:05:00:00	77.254.42.34	58.39.68.129	UDP	Brute Force	Medium	Blocked	1348	148.55	456	3024.3971344
7	01JAN24:06:00:00	112.108.176.251	120.11.51.145	ICMP	Brute Force	High	Blocked	1337	39.37	108	1231.8297223
8	01JAN24:07:00:00	3.173.150.154	128.116.142.157	ICMP	DDoS	Medium	Blocked	59	208.9	245	2358.4584673
9	01JAN24:08:00:00	116.115.120.25	135.138.128.252	ICMP	Malware	Medium	Blocked	278	61.9	255	1263.7804233
10	01JAN24:09:00:00	160.26.158.113	14.0.31.84	HTTPS	DDoS	Low	Blocked	1333	132.42	33	1783.3731307
11	01JAN24:10:00:00	181.27.237.216	148.55.187.208	HTTP	Phishing	Low	Blocked	457	19.59	23	525.5978661
12	01JAN24:11:00:00	149.198.170.29	38.210.66.109	ICMP	SQL Injection	Critical	Blocked	1373	247.61	333	3532.4772681
13	01JAN24:12:00:00	124.220.123.53	108.14.225.49	HTTPS	Brute Force	Medium	Successful	1094	37.87	54	929.15118178
14	01JAN24:13:00:00	202.172.86.30	123.99.116.60	UDP	Phishing	Medium	Blocked	1167	270.94	309	3587.3505411
15	01JAN24:14:00:00	88.80.186.159	1.141.22.69	HTTP	SQL Injection	Medium	Blocked	225	259.28	49	2433.7094404

```
*Identify data types;
proc contents data=SAS.cyber;
run;
```

The SAS System

The CONTENTS Procedure

Data Set Name	SAS.CYBER	Observations	1964
Member Type	DATA	Variables	14
Engine	V9	Indexes	0
Created	29/04/2025 09:48:14	Observation Length	136
Last Modified	29/04/2025 09:48:14	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	WINDOWS_64		
Encoding	wlatin1 Western (Windows)		

Engine/Host Dependent Information

Data Set Page Size	65536
Number of Data Set Pages	5
First Data Page	1
Max Obs per Page	481
Obs in First Data Page	462
Number of Data Set Repairs	0
ExtendObsCounter	YES
Filename	D:\sas\Sas project\cyber.sas7bdat
Release Created	9.0401M7
Host Created	X64_10HOME
Owner Name	RASHPINDER\hp
File Size	384KB
File Size (bytes)	393216

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
11	attack_score	Num	8	BEST12.	BEST32.
5	attack_type	Char	13	\$13.	\$13.
3	destination_ip	Char	15	\$15.	\$15.
9	duration	Num	8	BEST12.	BEST32.
14	log_attack_score	Num	8	BEST12.	BEST32.
13	log_duration	Num	8	BEST12.	BEST32.
10	num_packets	Num	8	BEST12.	BEST32.
8	packet_size	Num	8	BEST12.	BEST32.
4	protocol	Char	5	\$5.	\$5.
12	rand_noise	Num	8	BEST12.	BEST32.
6	severity	Char	8	\$8.	\$8.
2	source_ip	Char	15	\$15.	\$15.
7	status	Char	10	\$10.	\$10.
1	timestamp	Num	8	DATETIME.	ANYDTDTM40.

```

*Missing entries;
proc means data=SAS.cyber n nmiss;
run;

```

The SAS System

The MEANS Procedure

Variable	N	N Miss
timestamp	1945	19
packet_size	1948	16
duration	1964	0
num_packets	1940	24
attack_score	1964	0
rand_noise	1964	0
log_duration	1964	0
log_attack_score	1964	0

```
/*Identify and Count Duplicate Records */
proc sort data=SAS.cyber out=sorted_data nodupkey dupout=duplicates;
    by _all_;
run;
```

```
/* Count of Duplicate Records */
proc sql;
    select count(*) as duplicate_count
    from duplicates;
quit;
```

The SAS System	
duplicate_count	
	0

*UNIVARIATE ANALYSIS;

```
*Variable: duration;
proc means data=SAS.cyber;
var duration;
run;
```

```
proc univariate data=SAS.cyber;
var duration;
histogram/normal;
run;
```

```
proc sgplot data=SAS.cyber;
vbox duration;
run;
```

The SAS System

The MEANS Procedure

Analysis Variable : duration				
N	Mean	Std Dev	Minimum	Maximum
1964	149.8516421	90.1430034	0.1200000	573.2475162

The SAS System

The UNIVARIATE Procedure Variable: duration

Moments			
N	1964	Sum Weights	1964
Mean	149.851642	Sum Observations	294308.625
Std Deviation	90.1430034	Variance	8125.76106
Skewness	0.4944702	Kurtosis	0.94795325
Uncorrected SS	60053499.7	Corrected SS	15950869
Coeff Variation	60.1548319	Std Error Mean	2.0340484

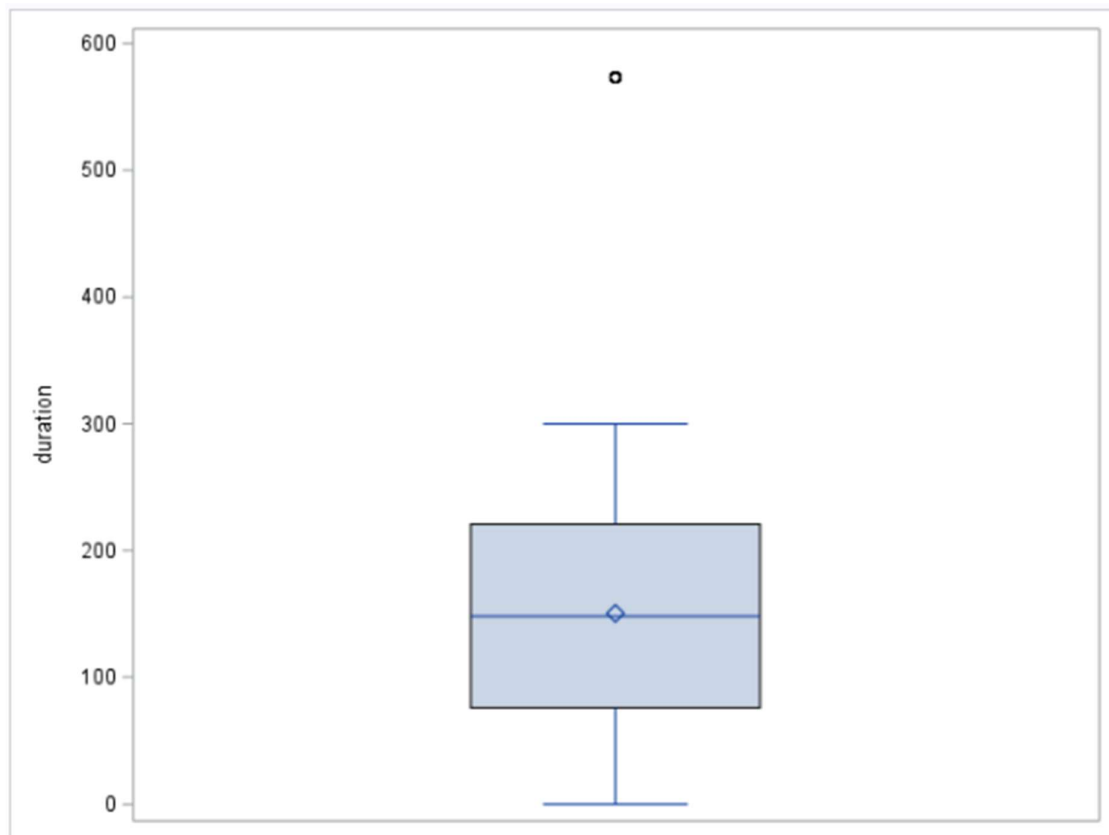
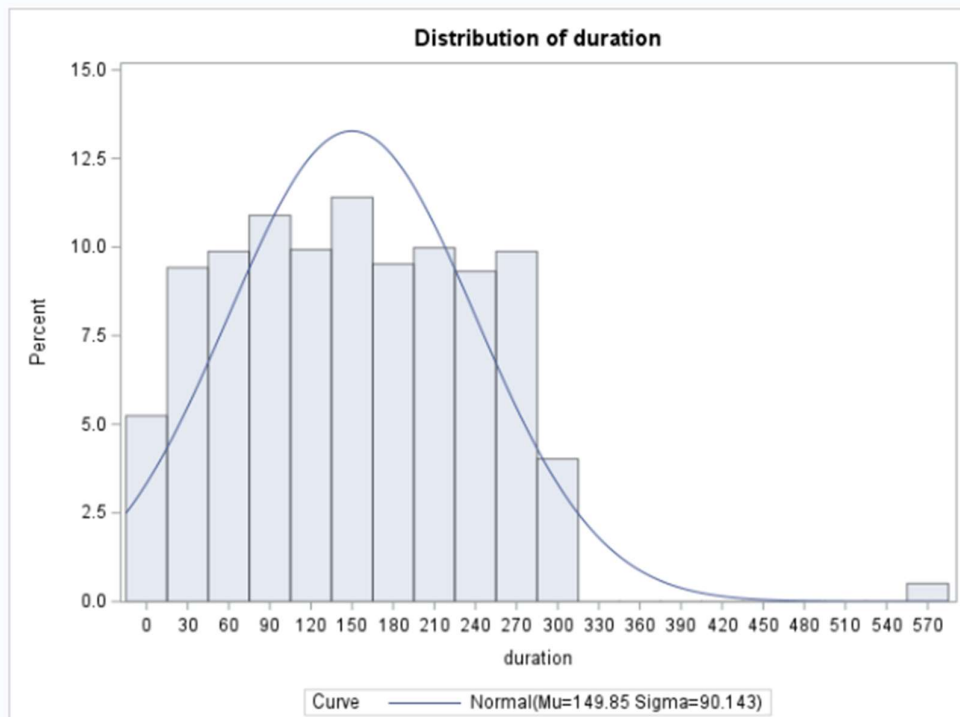
Basic Statistical Measures			
Location		Variability	
Mean	149.8516	Std Deviation	90.14300
Median	147.6100	Variance	8126
Mode	573.2475	Range	573.12752
		Interquartile Range	144.98500

Tests for Location: $\mu_0=0$				
Test	Statistic		p Value	
Student's t	t	73.67162	Pr > t	<.0001
Sign	M	982	Pr >= M	<.0001
Signed Rank	S	964815	Pr >= S	<.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	573.248
99%	298.290
95%	283.880
90%	269.240
75% Q3	220.820
50% Median	147.610
25% Q1	75.835
10%	31.240
5%	14.350
1%	2.870
0% Min	0.120

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
0.12	860	573.248	1488
0.17	1566	573.248	1491
0.22	191	573.248	1494
0.37	1107	573.248	1554
0.38	1705	573.248	1848

The UNIVARIATE Procedure



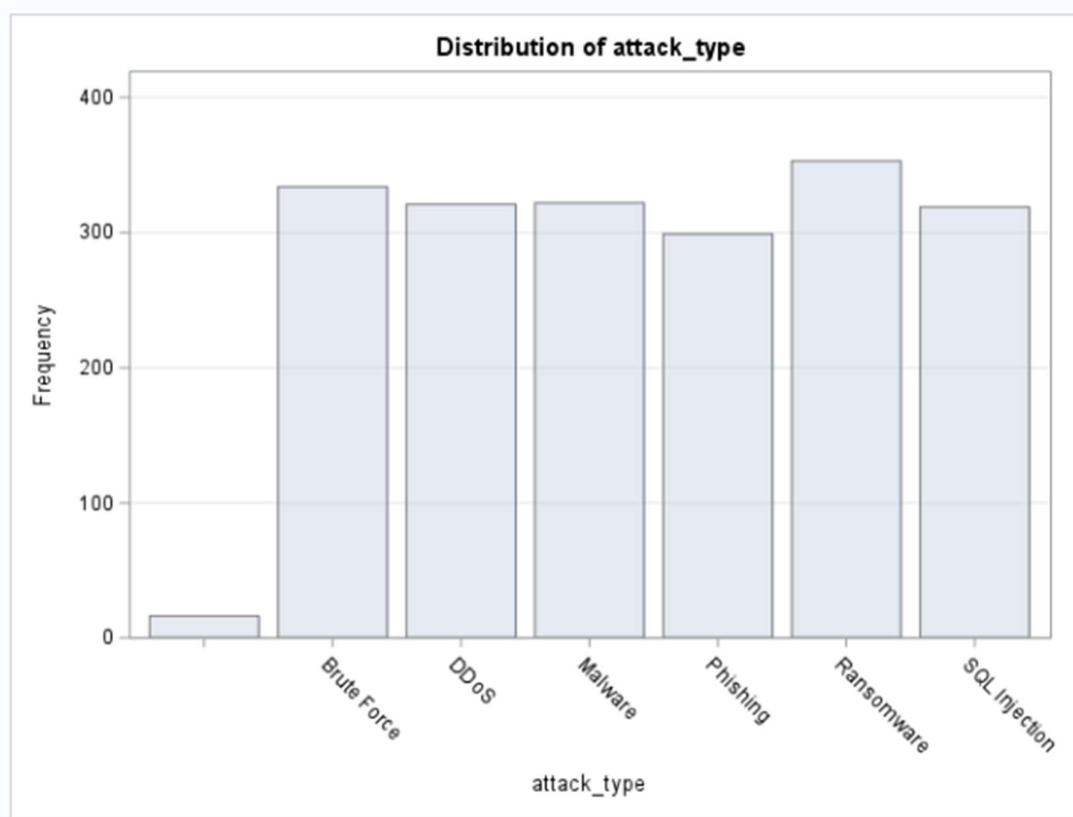
CATEGORICAL VARIABLE: PROC FREQ

```
proc freq data=SAS.cyber;  
table attack_type/plots=freqplot missing;  
run;
```

The SAS System

The FREQ Procedure

attack_type	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	16	0.81	16	0.81
Brute Force	334	17.01	350	17.82
DDoS	321	16.34	671	34.16
Malware	322	16.40	993	50.56
Phishing	299	15.22	1292	65.78
Ransomware	353	17.97	1645	83.76
SQL Injection	319	16.24	1964	100.00



```

*Outlier detection: Tukey's method;
%macro outlier_Analysis(data,var1,var2);
  proc univariate data=&data round=0.01;
    var &var1 &var2;
    output out=boxStats
    Qrange=iqr_var1 iqr_var2
    p75=Q3_var1 Q3_var2
    p25=Q1_var1 Q1_var2
    ;
  run;

  Data fence;
  set boxstats;
  *Tukey Method;;
  LIF_var1=Q1_var1-1.5*iqr_var1;*lower inner fence;
  UIF_var1=Q3_var1 +1.5 *iqr_var1;* upper inner fence;
/**/

/* *Extreme Tukey Method:*/
/* LOF_var1=Q1_var1-3*iqr_var1;*lower outer fence;*/
/* UOF_var1=Q3_var1 +3*iqr_var1;* upper outer fence;*/

  call symputx("LIF_var1",LIF_var1);
  call symputx("UIF_var1",UIF_var1);

  LIF_var2=Q1_var2-1.5*iqr_var2;*lower inner fence;
  UIF_var2=Q3_var2 +1.5 *iqr_var2;* upper inner fence;
  call symputx("LIF_var2",LIF_var2);
  call symputx("UIF_var2",UIF_var2);
  run;
  proc print data=fence;run;
  /*%put &LIF_var1 &UIF_var1 &LIF_var2 &UIF_var2;*/

  data detected_outliers;
  set &data;
  if &var1 < &LIF_var1 or &var1 > &UIF_var1 or &var2 < &LIF_var2 or &var2 > &UIF_var2 ;
  run;
  proc print data=detected_outliers; run;
%mend outlier_Analysis;


---


OPTIONS MPRINT;
%outlier_Analysis(SAS.cyber,duration,num_packets);

*Test of independency;;
*****
/*****CHI SQUARE*****/
*****;

```


Statistics for Table of attack_type by severity

Statistic	DF	Value	Prob
Chi-Square	15	24.0457	0.0643
Likelihood Ratio Chi-Square	15	24.2735	0.0606
Mantel-Haenszel Chi-Square	1	1.9549	0.1621
Phi Coefficient		0.1117	
Contingency Coefficient		0.1110	
Cramer's V		0.0645	

Sample Size = 1926
Frequency Missing = 38

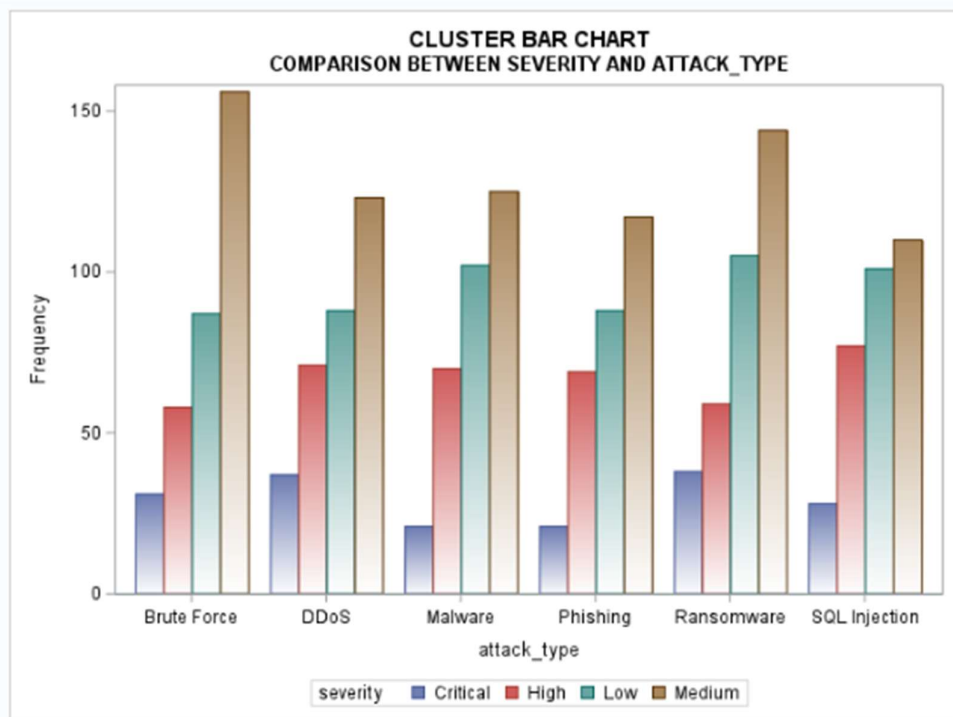
*

There is no statistically significant association between the variables (p-values > 0.05).

The strength of association is very weak, as shown by low values of Phi, Contingency Coefficient, and Cramer's V.

;

```
TITLE1 "CLUSTER BAR CHART ";
TITLE2 "COMPARISON BETWEEN SEVERITY AND ATTACK_TYPE ";
PROC SGPLOT DATA = SAS.cyber;
VBAR ATTACK_TYPE / GROUP = severity GROUPDISPLAY=CLUSTER FILLTYPE = GRADIENT;
RUN;
```



```

1
/* CATEGORICAL V/S CONTINUOUS */
/*Checking relation between Attack_type and Attack_score
Attack_type and Attack_score-CONTINUOUS VS CATEGORICAL-T-TEST

```

Assumptions for T-test:

- 1.Observations must be independent
- 2.Data in each group should be normally distributed
- 3.Variances of the two groups should be equal

TEST HYPOTHESIS

H0: The mean Attack Score is equal across Attack Type.

H1: The mean Attack Score is not equal across Attack Type.

```

*/

```

```

/*Checking normality in each group*/

```

```

/*We can rely on CLT because we have more than 30 onservations.*/

```

```

proc univariate data=SAS.cyber;
var attack_score;
histogram/normal;
run;

proc glm data=SAS.cyber;
class status;
model attack_score = status; * Predict attack_score based on status;
means status / hovtest=levене(type=abs) welch; * Test for homogeneity of variance and perfor
run;

TITLE1 'GROUP BOX PLOT';
TITLE2 'COMPARISON BETWEEN Status AND Attack-Score';
PROC SGPLOT DATA = SAS.cyber;
VBOX attack_score / GROUP = status;
RUN;

TITLE "BIVARIATE ANALYSIS OF CATEGORICAL V/S CONTINUOUS";
TITLE1 "Status AND Attack Type";
PROC TTEST DATA = SAS.cyber;
CLASS status;
VAR attack_score;
RUN;

```

Status AND Attack Type

The TTEST Procedure

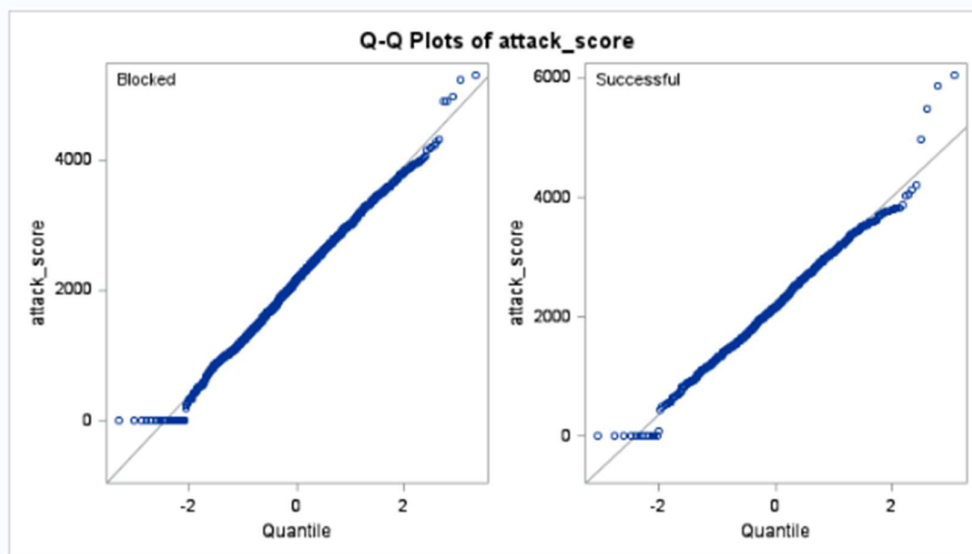
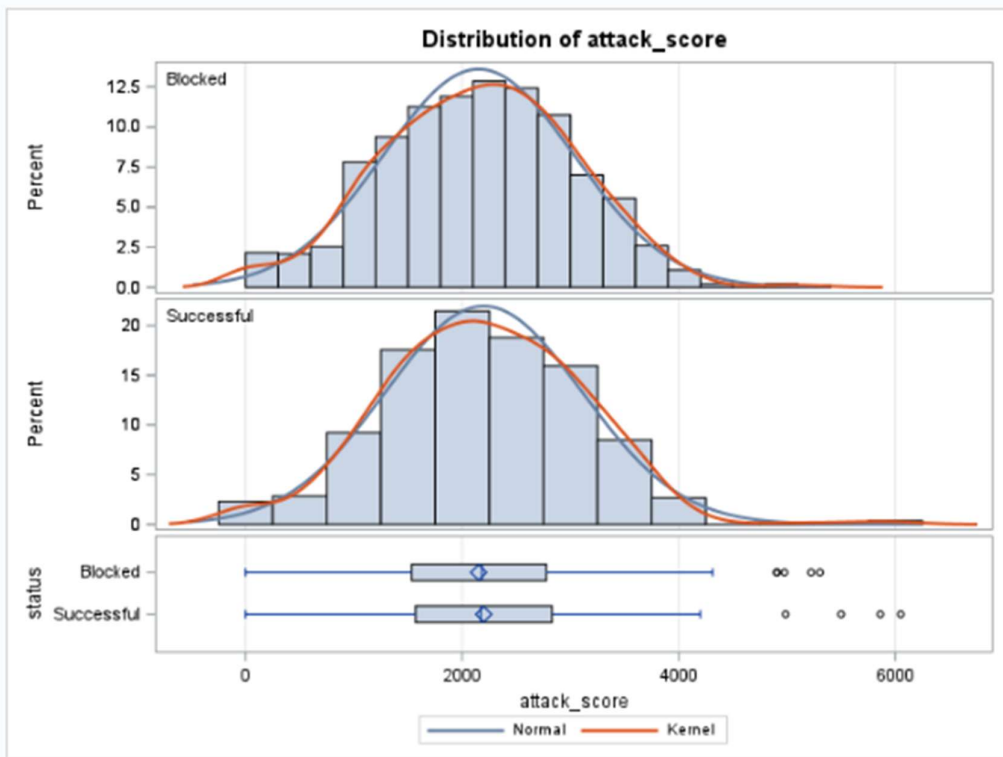
Variable: attack_score

status	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Blocked		1385	2152.3	879.8	23.6415	0	5304.0
Successful		564	2198.8	908.3	38.2467	0	6050.1
Diff (1-2)	Pooled		-46.5081	888.2	44.3643		
Diff (1-2)	Satterthwaite		-46.5081		44.9637		

status	Method	Mean	95% CL Mean	Std Dev	95% CL Std Dev
Blocked		2152.3	2105.9 2198.7	879.8	848.2 913.9
Successful		2198.8	2123.7 2273.9	908.3	858.2 964.7
Diff (1-2)	Pooled	-46.5081	-133.5 40.4983	888.2	861.1 917.0
Diff (1-2)	Satterthwaite	-46.5081	-134.7 41.7242		

Method	Variances	DF	t Value	Pr > t
Pooled	Equal	1947	-1.05	0.2946
Satterthwaite	Unequal	1015.1	-1.03	0.3012

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	563	1384	1.07	0.3603



```

/* CONTINUOUS V/S CONTINUOUS */
/*Assumptions of Pearson Correlation
  1.Linearity
  2.Homosidasticity
  3.Independence of observations
  4.No outliers

H0:There is no correlation between Packet_Size and Num_Packets.
H1:There is correlation between Packet_Size and Num_Packets.
*/

proc sgplot data=SAS.cyber;
  scatter x=attack_score y=duration;
  reg x=attack_score y=duration; /* Adds a trend line */
run;

title ' Correlation Matrix and Scatter Plot ';
proc corr data = SAS.cyber pearson spearman kendall
plots(maxpoint=none) = matrix(histogram);
var packet_size num_packets;
run;
title;

```

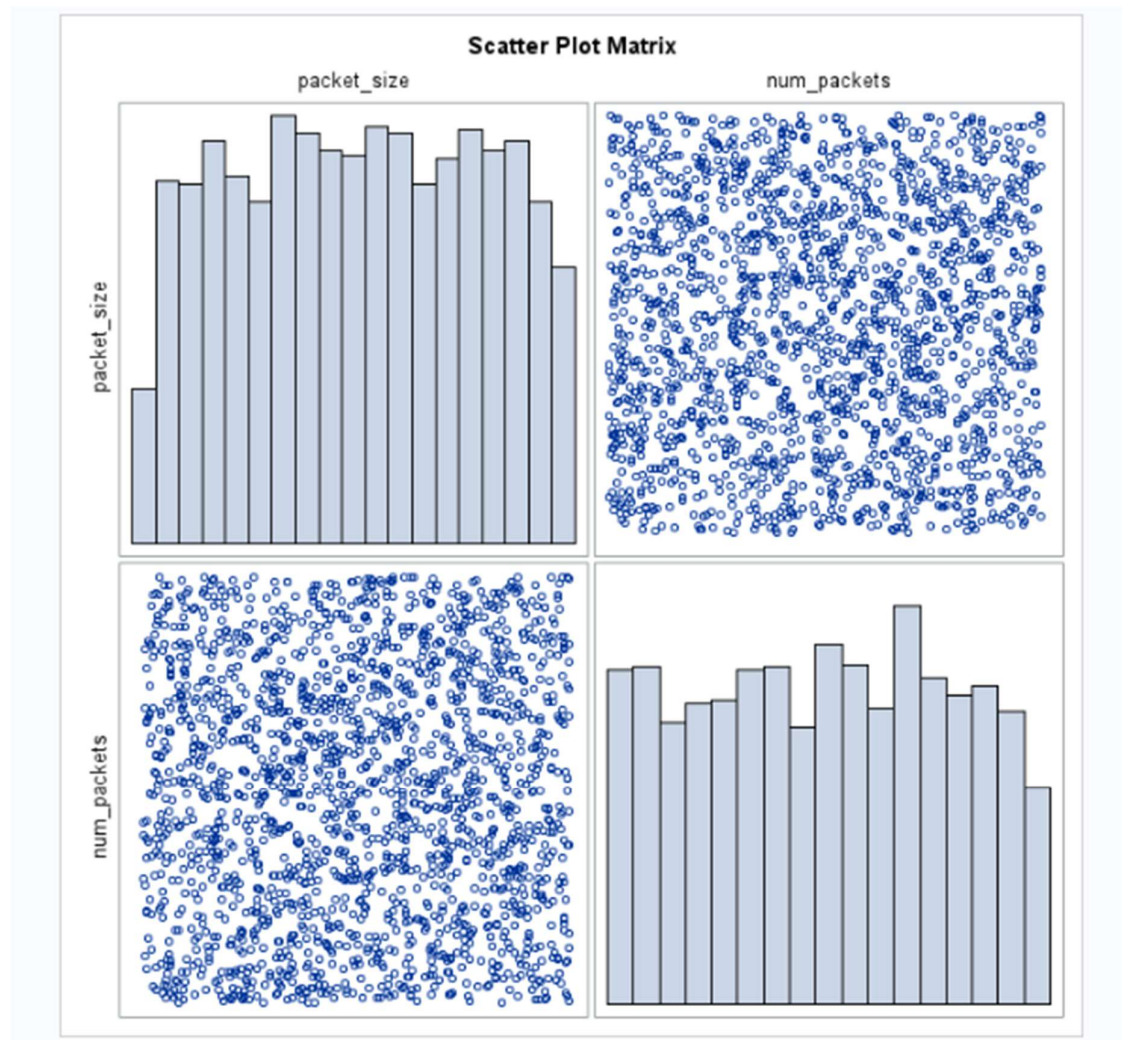
The CORR Procedure

2 Variables: packet_size num_packets

Simple Statistics						
Variable	N	Mean	Std Dev	Median	Minimum	Maximum
packet_size	1948	775.17094	414.56410	776.00000	40.00000	1499
num_packets	1940	250.17629	143.22023	253.00000	1.00000	499.00000

Pearson Correlation Coefficients Prob > r under H0: Rho=0 Number of Observations		
	packet_size	num_packets
packet_size	1.00000	0.01123
		0.6224
	1948	1925
num_packets	0.01123	1.00000
	0.6224	
	1925	1940

Spearman Correlation Coefficients Prob > r under H0: Rho=0 Number of Observations		
	packet_size	num_packets
packet_size	1.00000	0.01148
		0.6148
	1948	1925
num_packets	0.01148	1.00000
	0.6148	
	1925	1940



All the correlation coefficients are very close to 0, meaning almost no linear or monotonic relationship between `packet_size` and `num_packets`.

The p-values are all > 0.05 , indicating no statistically significant correlation.

Summary:

There is no meaningful or significant relationship between `packet_size` and `num_packets` based on Pearson, Spearman, and Kendall Tau correlations.

```
*Linearity;
proc sgplot data=SAS.cyber;
scatter x=attack_score y=duration;
run;
```

```
*Multicollinearity;
proc reg data=SAS.cyber;
model attack_score = packet_size duration num_packets / vif;
run;
```

```
*Residual Normality;
title "Linear Regression: Checking Residuals for Linearity";
proc reg data=SAS.cyber plots(only)=(residualbypredicted residualplot);
model attack_score = duration;
output out=SAS.reg_out predicted=pred residual=resid;
run;
```

*Most points are tightly spread around the horizontal line at 0, which is good.

However, there's a "fan-shaped" pattern with some downward residuals (below -2000), especially for higher predicted values.

This suggests a slight non-linearity or heteroscedasticity – variance of residuals increases with the predicted value.;

```
proc glm data=SAS.cyber;
model attack_score= duration; * Predict attack_score and duration without status;
means duration/ hovtest=levne(type=abs) welch; * Test for homogeneity of variance
and perform Welch's test if needed;
run;
```

```
proc reg data=SAS.cyber;
model attack_score = duration;
run;
```

Linear Regression: Checking Residuals for Linearity

The REG Procedure
Model: MODEL1
Dependent Variable: attack_score

Number of Observations Read	1964
Number of Observations Used	1964

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	945597377	945597377	3073.38	<.0001
Error	1962	603655651	307674		
Corrected Total	1963	1549253027			

Root MSE	554.68335	R-Square	0.6104
Dependent Mean	2163.10854	Adj R-Sq	0.6102
Coeff Var	25.64288		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1009.32999	24.28574	41.56	<.0001
duration	1	7.69947	0.13888	55.44	<.0001

Fit Diagnostics for attack_score

